

Urban Traffic Congestion Prediction Using CNN-LSTM Hybrid Deep Learning
ON

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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Name of Title: Urban Traffic Congestion Prediction Using CNN-LSTM Hybrid Deep Learning

Abstract:

Traffic congestion is a major problem in cities due to the increasing number of vehicles and rapid urban growth. It causes longer travel times, fuel wastage, air pollution, and stress for commuters. Predicting traffic congestion in advance helps people choose better routes and enables traffic authorities to manage roads more efficiently. In this project, a **CNN-LSTM Hybrid Deep Learning** model is used to predict urban traffic congestion. The **CNN (Convolutional Neural Network)** extracts important patterns from traffic data, while the **LSTM (Long Short-Term Memory)** learns how traffic changes over time. By combining these two models, the system provides more accurate traffic predictions than traditional methods. Historical traffic data is used to train the model, and the predicted results can help reduce congestion, improve travel planning, and support the development of smart city transportation systems.

INTRODUCTION:

Traffic congestion has become a common problem in many cities because of the growing population and the increasing number of vehicles on the roads. It leads to delays, fuel consumption, air pollution, and economic losses. Therefore, predicting traffic conditions before congestion occurs is important for improving road safety and transportation efficiency.

Traditional traffic prediction methods often fail to handle the complex nature of traffic because they cannot effectively learn both road patterns and changes over time. Deep learning techniques have shown better performance in solving this problem.

This project uses a **CNN-LSTM Hybrid Deep Learning** model for traffic congestion prediction. The **CNN** is used to identify important spatial features from traffic data, while the **LSTM** is used to learn traffic patterns over time. Combining these two models improves the accuracy of traffic prediction.

The model is trained using historical traffic data such as traffic volume, vehicle speed, and road occupancy. The predicted results can help traffic management authorities control traffic flow and assist commuters in selecting less congested routes. This project contributes to the development of intelligent transportation systems and smart cities by providing an efficient and reliable traffic prediction solution.

SOFTWARE REQUIREMENTS:

Software

Operating System

Programming Language

IDE

Deep Learning Library

Python Libraries

Dataset

Web Browser

Purpose

Windows 11

Python 3

Visual Studio Code (VS Code)

TensorFlow and Keras

NumPy, Pandas, Matplotlib, Scikit-learn

Traffic dataset from Kaggle or any public source

Google Chrome



Signature of the Guide



Course Instructor